Compression Thresholds For Temporal Model Accuracy In Deepfake Detection

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Introduction

- Current academic literature largely operates on the implicit assumption that temporal deepfake detection systems inherently outperform spatial models.
- There is a lack of direct studies comparing how the efficacy of these models changes under the compression levels applied by social media platforms.

Objective

- Identify the compression levels where the inherent advantage of temporal analysis is lost.
- See what accounts for the variance in the critical points between temporal models, and the type of fake being detected.
- Figure out if retraining models on only compressed data actually improves this overall performance.

Methodology

1. Source videos from FaceForensics++. Re-encode them at different compression levels.
2. Run them through DeepfakeBench using all possible detectors to measure accuracy. This allows us to control for augmentation, preprocessing, etc.
3. Break the data down with the DF method and detector type serving as the two points of reference (Figure 1).
4. Retrain models on the compressed data, with the worst performing CRF levels, to compare against the originals.
5. Visualize the data in Python with Matplotlib / Seaborn

Figure 1: Accuracy Degradation by Deepfake Method (CRF 0 to 40)

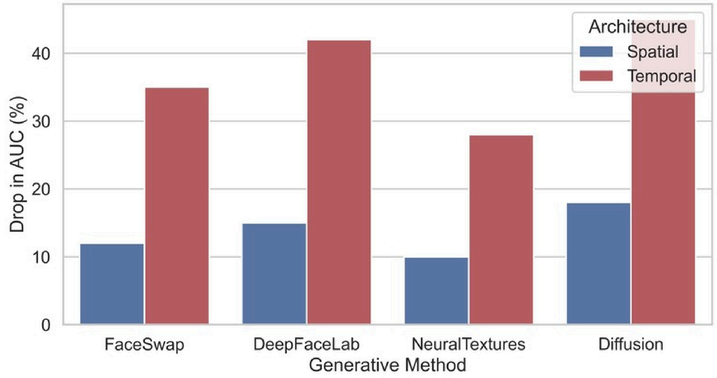


Figure 2: Performance Decay Under H.264 Compression

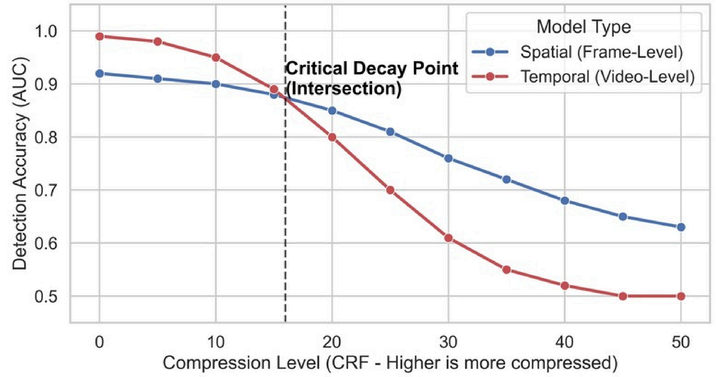


Figure 3: Temporal Architecture Robustness

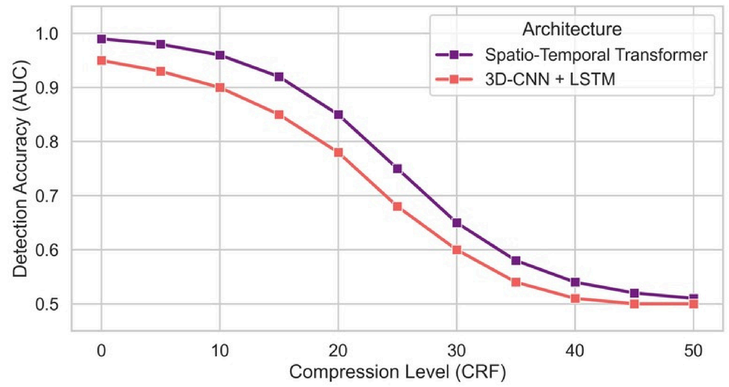
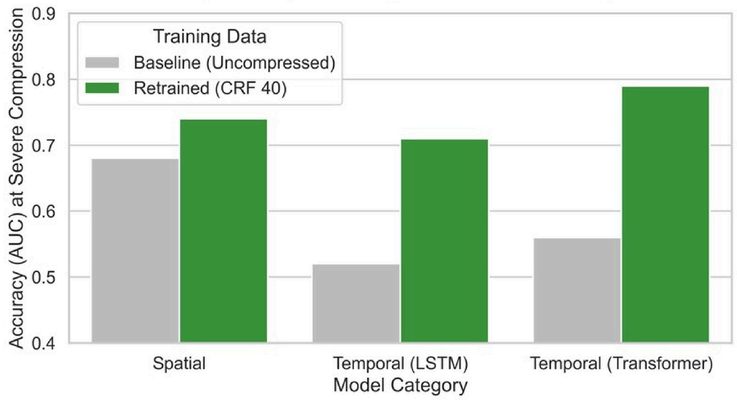


Figure 4: Impact of Compression-Aware Retraining



Expected Results

- We expect that video-level detection outperforms frame-level detection under low or no compression at all.
- We hope to show that there is a threshold below which temporal features cease to contribute measurably, and below it, video-level detection under performs against frame-level detection.
- We acknowledge that the threshold can vary based on the method of deepfake generation.
- We expect that transformer based models will perform better than LSTM, but that their retrained compression thresholds will not differ substantially.

Implications

- This data could provide the necessary weight distributions between temporal and spatial features to improve accuracy.
- The degradation thresholds of certain models may provide a basis for determining the best system for high compression environments.
- May show how the difference in permanent data loss differs between various models.